

PM Controller

Phase 0 — Action plan, milestones and build package · **tablet first**

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Backend Supabase Postgres **Issued** 20 September 2026

1. What changed, and what did not

The original package specified a React mobile-first PWA. The client is now a **native iPad app**, with the phone layout following from the same binary. Everything underneath is unchanged — the schema, the row-level security, and all nine non-negotiable principles never depended on the client.

REQUIREMENT FROM THE PHASE 0 RISK LIST	PWA ON IPAD / IPHONE	NATIVE
Rework notification must reach the Lead with the app closed	Fragile — needs Home Screen install first	APNs. Reliable.
Close-out under 60 seconds, camera one tap	File-picker detour	Direct camera
A close-out with no signal must not be lost	Browser storage evicted under pressure	SwiftData, survives
Crews shoot 12 MP images all day	JavaScript-side compression	Hardware pipeline

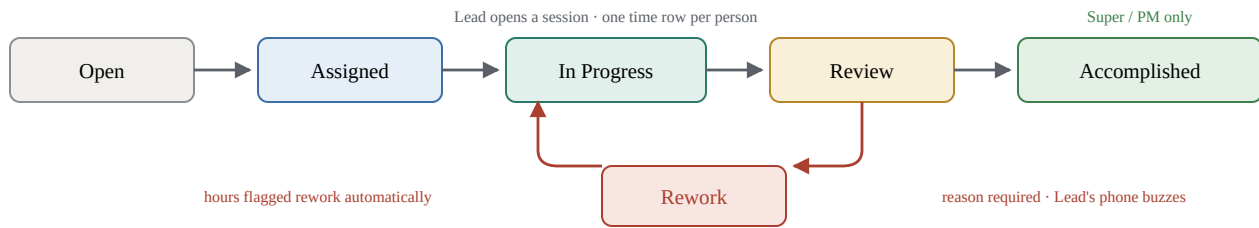
The cost, stated plainly. No Android, and App Store review on every release. Distribution to the Leads during the pilot is TestFlight, not the public store — faster, but capped at 100 internal testers with each build expiring after 90 days.

What it does *not* cost — seeing it work before paying for anything

WHAT YOU WANT TO DO	WHAT IT ACTUALLY COSTS
Compile it and run it in a Simulator	A Mac. No Apple account at all.
Install it on an iPad you own	A Mac and a free Apple ID. Re-sign every 7 days.
Put it on someone else's iPad (TestFlight)	\$99/yr Apple Developer Program
App Store	\$99/yr

And without a Mac at all. GitHub's macOS runners are free on public repositories, so CI compiles the app, boots an iPad simulator, launches it as each role, and uploads photographs of every screen as an artifact of every run. You can look at the real app before buying a thing. What that cannot give you is live iteration — clicking through and changing something on the spot. **The genuine prerequisite is a Mac, not the \$99.**

2. Status flow — unchanged, and locked in the database



Six core statuses, enforced by a Postgres enum. A seventh cannot be inserted from any client. Custom board columns sit on top and each maps to exactly one of these, so reporting reads core status and never a column name.

3. Continuous integration

Three jobs on every pull request. The first is the one that matters most: SPEC principle 8 says access control lives in the database, and this is what stops that claim quietly decaying as later phases add tables.

JOB	RUNNER	WHAT IT PROVES
Row-level security	ubuntu	Both migrations applied to a real Postgres 16, then all 56 assertions. Non-zero exit on regression.
Build for iPad	macOS	The Swift compiles. Nobody needs a Mac for this to be true.
Screenshots on iPad	macOS	The app runs, photographed as each role, downloadable from the run.

4. Phase plan and milestones

#	DELIVERABLE	MILESTONE — CHECKED WHEN	STATE
0	Plan, model, flow, checklist	This package issued	DONE
1	Schema · auth · admin user setup · row-level security	Admin creates a Lead by email; that Lead sees only his project and cannot read the receipts tables at all	DATABASE GATE MET 56/56 assertions green. iOS shell awaiting a Mac build.
2	Task board, columns, role-gated transitions	A Lead cannot move a task to Accomplished	NEXT
3	Time capture — crew selection, per-person rows	One session with 3 crew writes 3 rows	—
4	Photos, description, quantity at close-out	Task closes with a photo from a phone in the field	—
5	Review gate · Rework column · push	Rejected task lands in Rework and the Lead's phone buzzes	—
6a	Cost class, change orders, custom columns	CO hours excluded from base earned value	—
6b	Schedule fields, GC baselines, blocked periods	Blocked days attribute to a named party	—

6c	Work class, discovered defects, source links	A defect ticket links back to an Accomplished task	—
6d	Material page — upload, extraction, categories	A photographed receipt lands categorised, duplicates caught	—
7	Progress page, every number drillable	Clicking any figure returns its ticket list; numbers reconcile to a hand-check of one week	—
8	Pilot on a fresh PCM job	Two full weeks run without paper backup	—
9	Closeout — results, improvements, business case	Final PDF issued	—

5. Phase 1 — what is built and proven

The Phase 1 milestone is a database claim, and it is now tested rather than asserted. Fifty-six assertions run as the role a phone actually holds — `authenticated` , with a real user id — not as the database owner, who bypasses every policy. The suite exits non-zero when a rule regresses, so it gates CI.

The matrix, proven

AREA	TESTS	WHAT IS ACTUALLY ASSERTED
Project isolation	9	A Lead sees exactly one project. A sub's Lead on another job sees none of the first job's tasks. A Manager sees every project. The same man is Foreman on one job and Super on another.
Material access	15	Lead reads zero receipts, zero line items, zero categories. A Foreman reads his own receipt and cannot reach the other Foreman's by id. PM, Assistant PM and Manager read everything.
Sign-off	4	Lead, Foreman and Assistant PM are all refused. Super is allowed.
History	7	An Accomplished task cannot be reopened, by anyone, including the Manager. <code>task_events</code> cannot be rewritten even by the service role.
Change orders	4	Assistant PM and Super refused in both directions. PM allowed.
Locked statuses	5	A seventh status is impossible. A column cannot map to a status it does not match. <code>rework_count</code> cannot be typed, only derived.
Lead movement	4	A Lead moves his own task and cannot touch another Lead's, create a task, or kick work back.
People & cost	8	A Lead cannot read <code>burdened_rate</code> at all. Nobody can be deleted — deactivation is the only path.

Six defects found and fixed in the reviewed schema

Four of them would have stopped the app working entirely. Full write-up in `docs/DECISIONS.md` .

REF	DEFECT	EFFECT IF SHIPPED
FIX-1	<code>citext</code> created after the table that uses it	Migration could not apply at all
FIX-3	<code>auth.uid()</code> is null for service-role writes, but <code>actor_id</code> is NOT NULL	Every seed and background status change fails
FIX-4	The rework counter trips the guard that protects its own column	No kick-back can ever be recorded
FIX-5	Cost view unusable as <code>security_invoker</code> under column grants	Nobody sees burdened rates, PM included

FIX-6	The audit trigger is blocked by its own table's RLS	No status change saves, for any role, with no error that points at the cause
FIX-2	qty_needs_unit ignored qty_installed	Close-out records "340" with no unit

6. Reminders & checklist

Before any more code — these block the build

- ☒ Name the app and lock it. PM Controller. Bundle identifier `com.pmcontroller.app`.
- ☒ Separate it from Apex. An independent product; no contractor's name anywhere in the repo or the demo data.
- ☐ Answer: is there a Mac available? This is the real gate, not the \$99. Without one, CI screenshots are the only way to see the app.
- ☐ Create the Supabase project (free tier is sufficient for the pilot) and apply `0001`, `0002`, then seed.
- ☐ Confirm which iPads and iPhones the crews actually carry. iPadOS 17 is assumed; an iPad 5 drops the target to 16.
- ☐ Apple Developer Program enrolment (\$99/yr) — only when you are ready to put builds on other people's devices. Not before.

Before the pilot goes live

- ☐ Written OK from PCM ownership before any PCM data lives in a system you own.
- ☐ Decide data ownership up front: who owns the records if the relationship ends.
- ☐ Decide burdened rates: real numbers, or placeholder multipliers until PCM shares them.
- ☐ Confirm the area / system list matches how PCM actually breaks out its estimates.
- ☐ Confirm the material category list against PCM's estimate breakdown.
- ☐ Agree the root-cause list for defects — short and fixed, never free text.
- ☐ Get the GC's published schedule in a usable form, or accept manual milestone entry per area.
- ☐ TestFlight build in every Lead's hands, with notifications actually tested on their phone.
- ☐ 15-minute walkthrough with each Lead — one screen, one rule: close the task before you leave.
- ☐ Run one week in parallel with existing daily reports and compare totals.
- ☐ CSV export path confirmed out of every table, so nothing is trapped.

Standing rules during the build

- ☒ Core statuses locked in the database, not hidden in the UI.
- ☒ Every status change writes to `task_events`, no exceptions.
- ☒ Never allow direct edit of hours; only start/end times and crew selection.
- ☒ Row-level security enforced in the database, proven by test, not by the UI.
- ☒ Metrics built as saved filters from the start, so drill-down is free rather than a rewrite.
- ☒ CI gates the access rules, so a later phase cannot quietly break them.
- ☐ Tablet layout first; the phone layout follows by size class from the same binary.

- ☐ Offline tolerance: a close-out with no signal must not be lost.
- ☐ Photo compression on upload — receipts must stay legible enough to audit against.
- ☐ Rework notification must reach the Lead with the app closed.

7. Risks still live

RISK	HANDLING
Leads don't close tasks in the field	Close-out under 60 seconds: crew already selected, camera one tap, quantity one field. Measured on a real phone in Phase 4, not assumed.
PCM says "that's just part of your job"	PM Controller is an independent product, built on your own time and infrastructure. PCM is the pilot customer, not the owner. In writing before go-live.
Rework becomes a blame tool	Reasons describe the condition, not the person. Defect rate is never shown per individual — by system, area and phase only. This is built in, not policy.
Scope creep into a full ERP	v1 ships with no scheduling, no invoicing, no payroll export, no Android.
App Store review delays a fix	TestFlight for the pilot; expedited review exists for genuine blockers. Budget 24–48 hours for any store release.
Data trust	Phase 7 is not done until a week of app numbers matches a hand tally.

Next artifact: Phase 1 completion PDF — full chat interactions plus summary — issued once the app has been built green in CI and a real Lead has signed in against a live Supabase project. The database half of Phase 1 is already proven and is recorded above.